

OBJECT-ORIENTED SOFTWARE ENGINEERING

ASSIGNMENT 2 – DETAILED STEP-BY-STEP CALCULATIONS

Project: Smart Waste Management System

Based on Assignment 1 UML solution diagrams and Unit 2 estimation and risk-management formulas

Contents: Lorenz–Kidd Estimation • Use Case Points • Class Points • OOF • Comparison • Risk Exposure • RMMM Plan

Student Name: _____

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0. Common Assumptions Used in Calculations

The numerical classifications below are project-specific analytical assumptions made from the Assignment 1 Smart Waste Management System diagrams. The formulas, method structure and terminology follow the Unit 2 material.

Item	Assumption
Working time	1 person-day = 8 person-hours
Working month	1 person-month = 20 person-days
Lorenz–Kidd interface	Complex GUI, because the system includes dashboards, forms, GPS tracking, image upload and multiple role-based workflows
Lorenz–Kidd productivity	18 person-days per class
UCP productivity	20 person-hours per UCP
Class Point productivity	6 person-hours per adjusted class point
OOFp productivity	8 person-hours per adjusted function point
Adjustment ratings	Technical, environmental and GSC ratings are explicitly shown as analytical assumptions

1. Lorenz–Kidd Estimation – Step-by-Step Calculation

This method first estimates key/domain classes, then estimates support classes using an interface multiplier, and finally converts total classes into effort.

Step 1: Identify Key Classes

From the Assignment 1 class model and problem domain, the following 18 classes are treated as key/domain classes:

No.	Key Class
1	User
2	Citizen
3	MunicipalOfficer
4	Driver
5	Worker
6	Supervisor
7	Administrator
8	WasteBin
9	WasteReport
10	WasteClassification
11	CollectionSchedule
12	CollectionRequest
13	Vehicle
14	GPSLocation
15	WorkerAssignment
16	Complaint
17	WasteCollection
18	DemandPrediction

Therefore: **K = 18 key classes**

Step 2: Select Interface Multiplier

Interface Type	Multiplier
No GUI	2.0
Text-based UI	2.25
GUI	2.5
Complex GUI	3.0

The Smart Waste Management System is classified as a **Complex GUI**. Therefore:

Interface Multiplier = 3.0

Step 3: Calculate Support Classes

Formula: Support Classes = Key Classes × Interface Multiplier

= 18 × 3.0

= **54 support classes**

Step 4: Calculate Total Classes

Formula: Total Classes = Key Classes + Support Classes

= 18 + 54

= **72 total classes**

Step 5: Select Productivity

Assumed productivity = **18 person-days per class.**

Step 6: Calculate Total Effort

Formula: Effort = Total Classes × Person-Days per Class

= 72 × 18

= **1,296 person-days**

Step 7: Convert Person-Days to Person-Hours

1 person-day = 8 hours

= 1,296 × 8

= **10,368 person-hours**

Step 8: Convert Person-Days to Person-Months

1 person-month = 20 person-days

= 1,296 ÷ 20

= **64.80 person-months**

Final Lorenz–Kidd Result	Value
Key Classes	18
Support Classes	54
Total Classes	72
Effort	1,296 person-days
Person-Hours	10,368
Person-Months	64.80

2. Use Case Point (UCP) – Step-by-Step Calculation

Main Formula: $UCP = (UAW + UUCW) \times TCF \times ECF$

Effort Formula: $Effort = UCP \times Productivity\ Factor$

Step 1: Calculate Unadjusted Actor Weight (UAW)

Actor	Classification	Weight	Calculation
Citizen	Complex human GUI actor	3	$1 \times 3 = 3$
Municipal Officer	Complex human GUI actor	3	$1 \times 3 = 3$
Driver	Complex human GUI actor	3	$1 \times 3 = 3$
Waste Worker	Complex human GUI actor	3	$1 \times 3 = 3$
Supervisor	Complex human GUI actor	3	$1 \times 3 = 3$
Administrator	Complex human GUI actor	3	$1 \times 3 = 3$
Computer Vision / ML	Simple external system/API actor	1	$1 \times 1 = 1$

Human actors = $6 \times 3 = 18$

External ML actor = $1 \times 1 = 1$

UAW = $18 + 1 = 19$

Step 2: Calculate Unadjusted Use Case Weight (UUCW)

Complexity	Number of Use Cases	Weight	Calculation	Subtotal
Simple	10	5	10×5	50
Average	14	10	14×10	140
Complex	8	15	8×15	120

UUCW = $50 + 140 + 120 = 310$

Step 3: Calculate Technical Complexity Factor (TCF)

Formula: $TCF = 0.6 + 0.01 \times \sum Ti$

Technical Factor	Weight	Rating	Weighted Value
Distributed system	2.0	5	$2.0 \times 5 = 10.0$
Response time / performance	1.0	4	$1.0 \times 4 = 4.0$
End-user efficiency	1.0	4	$1.0 \times 4 = 4.0$
Complex internal processing	1.0	5	$1.0 \times 5 = 5.0$
Reusable code target	1.0	2	$1.0 \times 2 = 2.0$
Easy to install	0.5	3	$0.5 \times 3 = 1.5$
Easy to use	0.5	4	$0.5 \times 4 = 2.0$
Portable	2.0	3	$2.0 \times 3 = 6.0$
Easy to change	1.0	4	$1.0 \times 4 = 4.0$
Concurrency	1.0	4	$1.0 \times 4 = 4.0$
Security	1.0	5	$1.0 \times 5 = 5.0$
Third-party/service integration	1.0	3	$1.0 \times 3 = 3.0$
Training needs	1.0	2	$1.0 \times 2 = 2.0$
Total $\sum Ti$			52.5

$$TCF = 0.6 + 0.01 \times 52.5$$

$$= 0.6 + 0.525$$

$$TCF = 1.125$$

Step 4: Calculate Environmental Complexity Factor (ECF)

Formula: $ECF = 1.4 - 0.03 \times \sum Ei$

Environmental Factor	Weight	Rating	Weighted Value
Familiarity with development process	1.5	3	$1.5 \times 3 = 4.5$
Application experience	0.5	3	$0.5 \times 3 = 1.5$
OO experience of team	1.0	4	$1.0 \times 4 = 4.0$
Lead analyst capability	0.5	4	$0.5 \times 4 = 2.0$
Team motivation	1.0	4	$1.0 \times 4 = 4.0$
Stable requirements	2.0	3	$2.0 \times 3 = 6.0$
Part-time staff	-1.0	2	$-1.0 \times 2 = -2.0$
Difficult programming language	-1.0	2	$-1.0 \times 2 = -2.0$
Total $\sum Ei$			18.0

$$\begin{aligned} \text{ECF} &= 1.4 - 0.03 \times 18 \\ &= 1.4 - 0.54 \\ \mathbf{\text{ECF} &= 0.86} \end{aligned}$$

Step 5: Calculate Final UCP

$$\text{UCP} = (\text{UAW} + \text{UUCW}) \times \text{TCF} \times \text{ECF}$$

$$= (19 + 310) \times 1.125 \times 0.86$$

$$= 329 \times 1.125 \times 0.86$$

$$= 370.125 \times 0.86$$

$$= \mathbf{318.31 \text{ UCP}}$$

Step 6: Calculate Effort

$$\text{Productivity} = 20 \text{ person-hours/UCP}$$

$$\text{Effort} = 318.31 \times 20$$

$$= \mathbf{6,366.20 \text{ person-hours}}$$

Step 7: Convert to Person-Days

$$= 6,366.20 \div 8$$

$$= \mathbf{795.78 \text{ person-days}}$$

Step 8: Convert to Person-Months

$$= 795.78 \div 20$$

$$= \mathbf{39.79 \text{ person-months}}$$

Final UCP Result	Value
UAW	19
UUCW	310
TCF	1.125
ECF	0.86
UCP	318.31
Person-Hours	6,366.20
Person-Days	795.78
Person-Months	39.79

3. Class Point Method – Step-by-Step Calculation

The classes are grouped into PDT (Problem Domain Type), HIT (Human Interaction Type), DMT (Data Management Type) and TMT (Task Management Type).

Step 1: Determine Number of Classes in Each Type and Complexity

Type	Low	Average	High
PDT	14	4	0
HIT	0	0	0
DMT	1	0	0
TMT	1	2	0

Step 2: Apply Class Point Weights

Type	Low Weight	Average Weight	High Weight
PDT	3	6	10
HIT	4	7	12
DMT	5	8	13
TMT	4	6	9

Step 3: Calculate PDT Contribution

Low PDT = $14 \times 3 = 42$

Average PDT = $4 \times 6 = 24$

High PDT = $0 \times 10 = 0$

Total PDT = $42 + 24 + 0 = 66$

Step 4: Calculate HIT Contribution

Low HIT = $0 \times 4 = 0$

Average HIT = $0 \times 7 = 0$

High HIT = $0 \times 12 = 0$

Total HIT = 0

Step 5: Calculate DMT Contribution

Low DMT = $1 \times 5 = 5$

Average DMT = $0 \times 8 = 0$

High DMT = $0 \times 13 = 0$

Total DMT = 5

Step 6: Calculate TMT Contribution

Low TMT = $1 \times 4 = 4$

Average TMT = $2 \times 6 = 12$

High TMT = $0 \times 9 = 0$

Total TMT = $4 + 12 = 16$

Step 7: Calculate Total Unadjusted Class Points (TUCP)

$$\begin{aligned}\text{TUCP} &= \text{PDT} + \text{HIT} + \text{DMT} + \text{TMT} \\ &= 66 + 0 + 5 + 16 \\ &= 87\end{aligned}$$

Step 8: Calculate Value Adjustment Factor (VAF)

Formula: $\text{VAF} = 0.65 + 0.01 \times \Sigma\text{GSC}$

Assumed $\Sigma\text{GSC} = 48$

$$\begin{aligned}\text{VAF} &= 0.65 + 0.01 \times 48 \\ &= 0.65 + 0.48 \\ &= 1.13\end{aligned}$$

Step 9: Calculate Adjusted Class Points

$$\begin{aligned}\text{CFP} &= \text{TUCP} \times \text{VAF} \\ &= 87 \times 1.13 \\ &= 98.31 \text{ adjusted class points}\end{aligned}$$

Step 10: Calculate Effort

Productivity = 6 person-hours/class point

$$\begin{aligned}\text{Effort} &= 98.31 \times 6 \\ &= 589.86 \text{ person-hours}\end{aligned}$$

Step 11: Convert to Person-Days

$$\begin{aligned}&= 589.86 \div 8 \\ &= 73.73 \text{ person-days}\end{aligned}$$

Step 12: Convert to Person-Months

$$\begin{aligned}&= 73.73 \div 20 \\ &= 3.69 \text{ person-months}\end{aligned}$$

Final Class Point Result	Value
TUCP	87
VAF	1.13
Adjusted Class Points	98.31
Person-Hours	589.86
Person-Days	73.73
Person-Months	3.69

4. Object-Oriented Function Point (OOFP) – Step-by-Step Calculation

Object-oriented elements are mapped to function-point categories: persistent classes → ILF, external interfaces → EIF, input transactions → EI, derived outputs → EO and retrievals → EQ.

Step 1: Calculate ILF

Identified Low ILFs = 13

Weight of Low ILF = 7

$$\begin{aligned}\text{ILF} &= 13 \times 7 \\ &= \mathbf{91}\end{aligned}$$

Step 2: Calculate EIF

Identified Low EIFs = 3

Weight of Low EIF = 7

$$\begin{aligned}\text{EIF} &= 3 \times 7 \\ &= \mathbf{21}\end{aligned}$$

Step 3: Calculate EI

Low EIs = $22 \times 3 = 66$

Average EIs = $6 \times 4 = 24$

$$\mathbf{EI = 66 + 24 = 90}$$

Step 4: Calculate EO

Low EOs = $4 \times 4 = 16$

Average EOs = $3 \times 5 = 15$

$$\mathbf{EO = 16 + 15 = 31}$$

Step 5: Calculate EQ

Low EQs = $5 \times 4 = 20$

Average EQs = $1 \times 5 = 5$

$$\mathbf{EQ = 20 + 5 = 25}$$

Step 6: Calculate Unadjusted Function Points (UFP)

$$\begin{aligned}\text{UFP} &= \text{ILF} + \text{EIF} + \text{EI} + \text{EO} + \text{EQ} \\ &= 91 + 21 + 90 + 31 + 25 \\ &= \mathbf{258}\end{aligned}$$

Step 7: Calculate Value Adjustment Factor (VAF)

$$\begin{aligned}\text{VAF} &= 0.65 + 0.01 \times \Sigma\text{GSC} \\ \Sigma\text{GSC} &= 48 \\ \text{VAF} &= 0.65 + 0.01 \times 48 \\ &= 0.65 + 0.48 \\ &= \mathbf{1.13}\end{aligned}$$

Step 8: Calculate Adjusted Function Points (AFP)

$$\begin{aligned}\text{AFP} &= \text{UFP} \times \text{VAF} \\ &= 258 \times 1.13 \\ &= \mathbf{291.54}\end{aligned}$$

Step 9: Calculate Effort

Productivity = 8 person-hours/function point

$$\begin{aligned}\text{Effort} &= 291.54 \times 8 \\ &= \mathbf{2,332.32 \text{ person-hours}}\end{aligned}$$

Step 10: Convert to Person-Days

$$\begin{aligned}&= 2,332.32 \div 8 \\ &= \mathbf{291.54 \text{ person-days}}\end{aligned}$$

Step 11: Convert to Person-Months

$$\begin{aligned}&= 291.54 \div 20 \\ &= \mathbf{14.58 \text{ person-months}}\end{aligned}$$

Final OOF Result	Value
ILF	91
EIF	21
EI	90
EO	31
EQ	25
UFP	258

Final OOFP Result	Value
VAF	1.13
AFP	291.54
Person-Hours	2,332.32
Person-Days	291.54
Person-Months	14.58

5. Comparison of All Estimation Methods

Method	Size Result	Person-Hours	Person-Days	Person-Months
Lorenz-Kidd	72 estimated classes	10,368.00	1,296.00	64.80
Use Case Point	318.31 UCP	6,366.20	795.78	39.79
Class Point	98.31 adjusted CP	589.86	73.73	3.69
OOFp	291.54 adjusted FP	2,332.32	291.54	14.58

Recommended primary method: **Use Case Point (UCP)**, because Assignment 1 already provides a detailed use-case diagram with actors and system functionality. OOFp is a useful secondary cross-check for transactional functionality.

6. Risk Analysis – Step-by-Step Calculation

Risk Exposure Formula: $RE = \text{Probability} \times \text{Consequence}$

Impact/consequence is expressed in estimated person-days. Each risk is calculated separately.

Risk	Probability (P)	Impact (C)	Calculation	Risk Exposure
R1 Requirements change	0.55	35	0.55×35	19.25
R2 Poor ML classification accuracy	0.40	45	0.40×45	18.00
R5 Service integration failure	0.35	50	0.35×50	17.50
R6 Security breach	0.25	60	0.25×60	15.00
R3 Inaccurate demand prediction	0.35	40	0.35×40	14.00
R11 Key specialist unavailable	0.25	50	0.25×50	12.50
R9 Vehicle/worker shortage	0.40	30	0.40×30	12.00
R12 Peak performance problems	0.30	40	0.30×40	12.00
R8 Workflow delays	0.45	25	0.45×25	11.25
R7 Database failure/data loss	0.20	55	0.20×55	11.00
R4 GPS outage/inaccuracy	0.30	30	0.30×30	9.00
R10 Poor network connectivity	0.45	20	0.45×20	9.00

Priority order is determined by sorting Risk Exposure from highest to lowest. Highest exposure: R1, followed by R2 and R5.

7. RMMM Plan – Risk Mitigation, Monitoring and Management

Risk	Mitigation	Monitoring	Management / Contingency
R1 Requirements change	Freeze baseline after approval; apply formal change control.	Review change requests weekly.	Use contingency buffer and re-estimate affected work.
R2 ML classification accuracy	Validate training data; define confidence threshold.	Track accuracy and misclassification rate.	Send low-confidence cases for manual verification.
R5 Integration failure	Define APIs/contracts early; run integration tests.	Track failed API calls and integration pass rate.	Use retries, queues and isolate the failing service.
R6 Security breach	RBAC, secure authentication, validation and audit logs.	Monitor failed logins and unusual access.	Disable affected accounts, investigate logs and restore secure state.
R3 Prediction inaccuracy	Validate model and allow human override.	Monitor prediction error.	Use manual/rule-based prioritization until corrected.
R11 Specialist unavailable	Document work and cross-train team members.	Review staffing availability.	Reassign work or use backup specialist.
R9 Resource shortage	Maintain availability and backup resource data.	Compare planned vs actual availability.	Re-optimize schedules and allocate backup resources.
R12 Performance problems	Load testing, queues and capacity planning.	Monitor response time and utilization.	Scale services and defer non-critical processing.

8. Final Submission Notes

For each estimation method, the calculation sequence in this PDF is: **(1)** identify items from Assignment 1, **(2)** classify them, **(3)** apply the required weights, **(4)** show multiplication, **(5)** add subtotals, **(6)** apply adjustment factors, **(7)** calculate effort, and **(8)** convert hours into person-days and person-months.

All project-specific ratings and classifications that are not explicitly printed in the UML diagrams are stated as assumptions and should be presented as such during submission or viva.